

DEVELOPMENT OF PLASMA SOURCE FOR SEED GERMINATION

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ABSTRACT

Plasma is one of five fundamental states of matter (the other four being solid, liquid, gas and Einstein Bose condensate) characterized by the presence of a significant portion of charged particles in any combination of ions or electrons.

This report emphasizes on harnessing innovative plasma technology to stimulate faster seed germination and cultivate healthier plants right at home. This method is cost-effective and accessible, perfect for home gardeners and small-scale farmers eager to increase their crop potential without expensive equipment.

Plasma treated seeds are an innovative approach toward increasing yield as it does not involve any use of chemicals or substances that alter the natural nutrients of the plant as well as benefit the environment and the user economically as plasma treated seeds require less water and prevent pollution caused due to excess usage of fertilizers and pesticides.

Throughout this report, we will observe and inform about the fascinating science behind plasma, explore practical plasma sources, and provide step-by-step instructions to create plasma seed enhancer to boost yields efficiently.

CHAPTER 1

INTRODUCTION

How can we use plasma to enhance the growth of plant from seed state?

Usage of artificial fertilizers or pesticides to avoid infestation and improve growth can prove to be hazardous if used out of proportions and leads to water and land pollution. This many a times this leaves the land unfertile and uncultivable after few cycles of yield.

Our objective is to use plasma to enhance the growth of seeds used in domestic purposes. Some implications of treatment of seeds with plasma result in following enhancements:

Surface Modification : Plasma treatment etches the seed coat, enhancing water uptake by making surfaces more hydrophilic, improving germination.

Pathogen Sterilization : It effectively reduces seed-borne pathogens like Fusarium by over 80%, lowering disease risk during early growth stages.

Metabolic Stimulation: The plasma's energetic particles activate seed metabolism, both enhancing germination speed and seedling vigor.

This process can enhance seed germination rates, speed up the germination process, and even rejuvenate older seeds. Plasma treatment can also decontaminate seeds, protect them against stress, and even enrich them with micronutrients

CHAPTER 2

LITERATURE SURVEY

Existing works in the field of plasma generation is summarized in the following pointers:

1. Paper by Benjamin K Barnes and Habilou Ouro-Koura in American journal of physics.
2. Generation and Applications of Plasma by Faizan Mehmood and Tariq Kamal.
3. Cold plasma technology: A cutting-edge approach for enhancing shrimp preservation by Hedayat Hosseini and Adel mirza.
4. Indigenous Cold Atmospheric Pressure Plasma System: 10 MHz (ICAPPS-10) by Bhabha Atomic Research Centre (govt. of India).

LEARNINGS:

Plasma finds applications in diverse fields, including medical sciences, manufacturing, energy production, and environmental cleanup. Plasma proves to be an efficient alternative to other methods for treatment and sterilization of bio hazardous material and various benefits in agriculture.

CHAPTER 3

SYSTEM ANALYSIS & DESIGN

Development of Plasma source for seed Germination consists of the following components:

3.1 Fly back Transformer:

The flyback transformer is a special class of transformer family. Fundamentally it's a step-up transformer, but with a huge potential of stepping up the voltage. As compared to power transformers, it is compact in size and mobile. One of the common applications of flyback transformers is in CRT tube televisions, where a very high voltage is required in the picture tube. For an input of 230 V, a flyback transformer can obtain an output of up to 20,000 V. Such is the potential of flyback transformers. It can even operate with a low voltage such as 12 V or 5V.

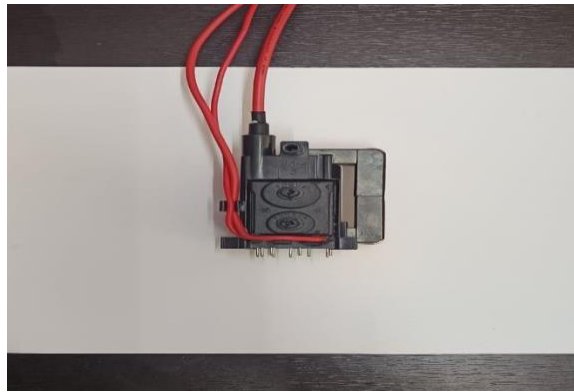


Fig 3.1 Flyback transformer

3.2 12 Volt dc adapter:

A 12V DC adapter is a device that converts AC power from a wall outlet into 12 volts of direct current (DC). These adapters are commonly used to power various electronic devices. The adapter's amperage (e.g., 2A, 5A) determines the maximum power it can supply to the connected device. It's crucial to match the adapter's voltage and amperage to the device's requirements to ensure proper function and prevent damage.



Fig 3.2 12v dc adopter

3.3 NPN bjt (NEC D363) and heat sink:

A heat sink is used with an NPN BJT (like the NEC D363) to dissipate excess heat generated by the transistor during operation, preventing overheating and ensuring reliable performance. The heat sink effectively conducts heat away from the transistor, transferring it to the surrounding environment. A heat sink is essential for power transistors like the NEC D363 to prevent overheating and ensure reliable operation. Choosing and using a suitable heat sink involves calculating the required thermal resistance and ensuring proper mounting.

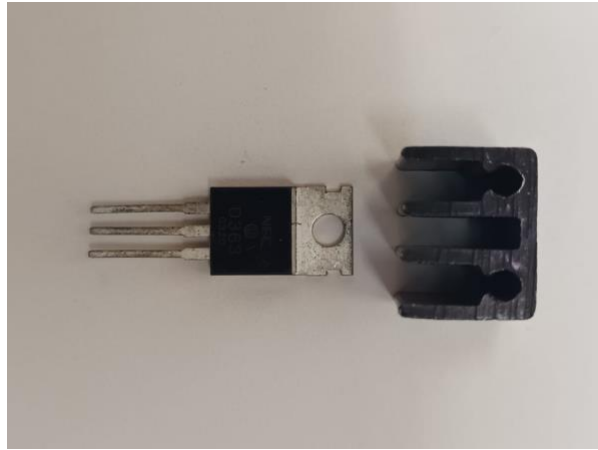


Fig 3.3 NPN bjt (NEC D363) and heat sink

3.4 Resistors and Perfboard:

A 220-ohm resistor is a common value, often a through-hole type with a carbon film construction. These resistors typically have a power rating of 0.25 watts. 10 Ohm resistors are readily available in various configurations. Common types include carbon film resistors with a 1/4 Watt power rating.

A perfboard, also known as a perforated circuit board, prototyping board, or dot PCB, is a sheet of insulating material with copper pads. Typically made of fiberglass or phenolic resin, they are used for prototyping and electronic projects. They provide a base for soldering electronic components.

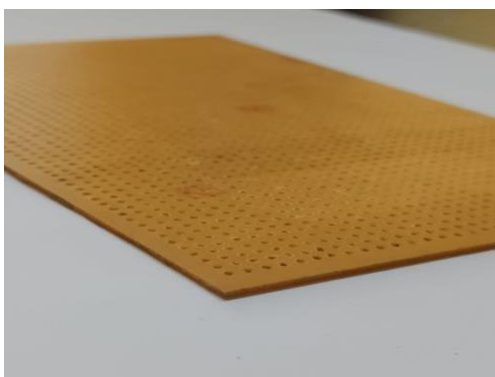


Fig 3.4 Perfboard

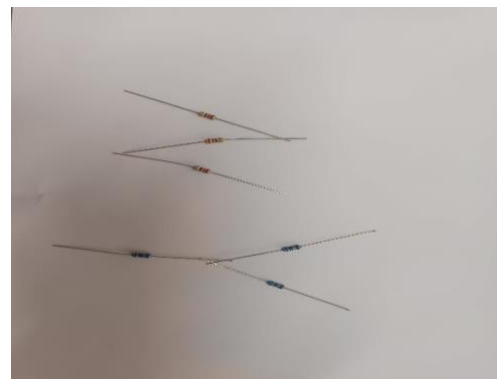


Fig 3.5 Resistor (220 and
10 ohm)

CHAPTER 4

IMPLEMENTATION

A Development of Plasma source for seed Germination works by ionizing a gas, creating a state of matter where a portion of the gas is electrically neutral and free electrons exist. This process is typically achieved by applying a high-frequency electric field or electromagnetic radiation to a gas. The energy from this field excites the gas, causing some of its atoms to lose electrons and become ionized. The resulting mixture of ions, electrons, and neutral particles is called plasma.

The working of it involves a power supply in this case 12 V battery which supplies the required power. Then the NPN BJT transistor converts direct current to pulsating direct current. This pulsating dc is then supplied to the flyback transformer whose main function is stepping up the voltage. Then the outlet to this flyback transformer is given to a metallic pin, which when kept at an optimal distance from the ground conducts plasma by ionizing the air between the metallic pin and ground. A step ahead we put multiple pin output and provide grounding to the dielectric which whose conduction of plasma is better. A plate of glass is used to provide a platform for the seed to be treated with plasma between the pin hex and the ground.

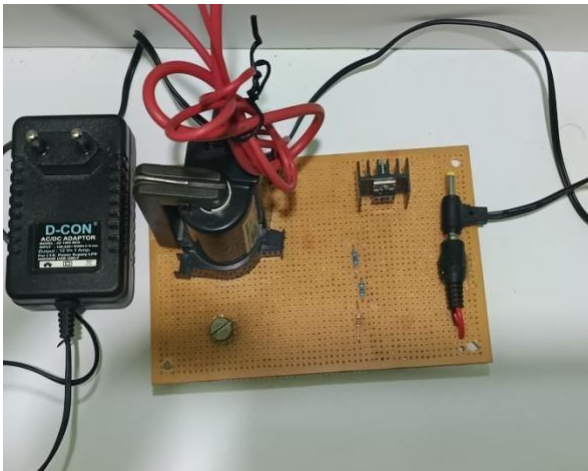


Fig 4.1 WORKING MODEL

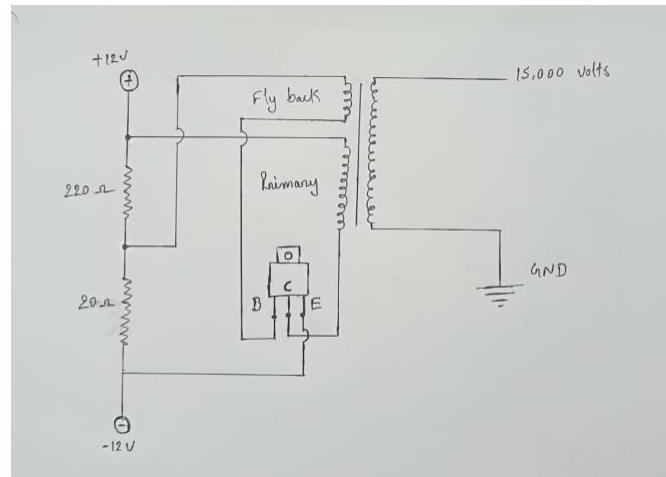


Fig 4.2 CIRCUIT DIAGRAM

CHAPTER 5

RESULTS AND DISCUSSION

The experiment is conducted successfully and the growth of plasma enhanced seed is compared to normal seed. We are able to see the stark difference as plasma enhanced seed required less water, was able to avoid infestation and had better growth.

Reports indicate that plasma treatment positively affects seed germination and seedling growth. Studies show increased germination percentages and, in some cases, improvements in plant characteristics like capitulum diameter and seed yield per capitulum. This technology is applied during different stages of agricultural food production to promote germination.

We were able to widen our perspective on the various application of plasma in multiple fields of work and gain the knowhow of the working of a Development of Plasma source for seed Germination.

CHAPTER 6

APPLICATIONS AND ADVANTAGES

Plasma enhanced seeds offer several advantages, including improved germination rates, faster germination, enhanced seedling growth, and increased tolerance to stress conditions like drought and salinity. This technology also helps with seed decontamination and can potentially rejuvenate old seeds, making it a promising alternative to traditional seed processing methods.

Enhanced Germination and Seedling Growth:

Improved Germination Rate: Plasma treatment can significantly increase the percentage of seeds that germinate, leading to a higher plant population. **Faster Germination Speed:**

Seeds treated with plasma tend to germinate more quickly, shortening the time it takes for seedlings to emerge.

Enhanced Seedling Growth: Plasma treatment can improve seedling characteristics like root and shoot development, leading to healthier and more vigorous plants. **Increased Water Uptake:**

Plasma treatment can enhance water uptake in seeds, which is crucial for germination and early seedling growth.

Decontamination and Seed Health:

Pathogen Inactivation: Plasma treatment can effectively kill pathogens and other microorganisms present on seed surfaces, reducing the risk of disease transmission. **Seed Decontamination:** this technology can be used to decontaminate seeds, ensuring that they are free from harmful pathogens and contaminants. **Rejuvenation of Old Seeds:** Plasma treatment can help to rejuvenate old seeds, extending their viability and germination potential.

Other Benefits: Plasma treatment is an environmentally friendly and sustainable method, as it does not involve the use of harmful chemicals. **Potential for Increased Yields:** By improving germination, seedling growth, and stress tolerance, plasma treatment can potentially lead to increased crop yields.

Other fields that find application of plasma are Plasma has diverse applications, particularly in medical fields. It is utilized in developing new medical treatments, such as wound healing, cancer therapy, and enhancing bone healing. The use of plasma in medicine is rapidly evolving, showing promise in various therapeutic areas. The ability of plasma to interact with biological tissues makes it a valuable tool for developing innovative treatments.

CHAPTER 7

CONCLUSION

The successful development of a plasma source for seed germination marks a significant advancement in sustainable agricultural technology. This project focused on creating a low-cost, efficient plasma generation system using a high-voltage flyback transformer capable of producing a cold atmospheric plasma environment suitable for seed treatment. The plasma source was tested for its ability to enhance seed germination by exposing seeds to reactive species such as ions, electrons, and radicals, which are known to activate physiological and biochemical changes within the seed.

The experimental results indicated that plasma treatment led to a noticeable improvement in germination rates, reduced germination time, and enhanced overall seedling vigor. This is primarily due to the plasma's ability to break seed dormancy, sterilize the seed surface, and increase water uptake by altering the seed coat's permeability. Moreover, the process is environmentally friendly and eliminates the need for chemical enhancers or pesticides, making it a sustainable and safe solution for modern agriculture.

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4. Set forth in the IEEE PSPB Operations Manual (sections 8.2.1.C & 8.2.2.A).
5. <https://www.elprocus.com>